THE GEOMETRY OF FISHER'S EARLY STATISTICS: A MODERN TREATMENT OF TWO FOUNDATIONAL PAPERS

Abstract. We give a modern geometric account of the mathematics in two early papers of R. A. Fisher: his 1920 comparison of the mean error and mean square error as estimators of scale (*Monthly Notices of the Royal Astronomical Society*), and his 1921 treatment of the sampling distribution of the correlation coefficient (*Metron*). Fisher's original arguments are recast in the language of Riemannian geometry and tensor notation: the Fisher information $g_{ab}(\theta)$ as a metric tensor on parameter space, the z-transformation as the geodesic coordinate in which the Christoffel symbols approximately vanish, and sufficiency as rotational invariance on the deviation sphere. The central narrative is that sample space carries a flat metric δ_{ij} whose orthogonal decompositions encode sufficient statistics, while parameter space carries a curved metric g_{ab} whose geodesic coordinates dictate the right parametrizations for inference.

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1. Introduction

In 1920 and 1921, Fisher published two papers that—beneath archaic notation and an astronomical motivation—contain ideas of striking modernity:

(i) **Fisher** [1] asks: given n observations from a normal population, is the sample standard deviation or the mean absolute deviation a better estimator of the scale parameter σ? Fisher answers this by introducing the *representative point* of a sample in $\mathbb{R}^n$, decomposing $\mathbb{R}^n$ into orthogonal subspaces, and proving what we now call *sufficiency* of the sample variance for the normal family.

(ii) **Fisher** [2] derives the exact sampling distribution of the correlation coefficient r from n bivariate normal pairs, and introduces the *z-transformation* $z = \operatorname{arctanh}(r)$ that normalizes and variance-stabilizes the distribution. The paper also contains Fisher's first published distinction between probability and likelihood.

Fisher's original presentations are brilliant but notoriously difficult: the notation is a century old, and pages of numerical tables and special-case calculations obscure the geometric ideas that drive the arguments. We retell both stories in a unified modern language, stripping away the computational scaffolding to expose the clean geometric narrative underneath. The central thread is the passage from the flat metric on sample space to the curved metric on parameter space.

Scope. This paper is a *geometric* commentary. Fisher's originals contain pages of numerical tables, eight worked examples, and detailed special-case calculations that served a concrete inferential purpose; we do not attempt to reproduce all of that material. We do include a small number of worked examples (section 6) to show how the geometry translates into practice, but readers seeking the full computational toolkit should consult the originals directly.

Index conventions. We use Latin indices $i, j, k = 1, \dots, n$ for coordinates on sample space $\mathbb{R}^n$ (with the Euclidean metric δ_{ij}) and early Latin indices $a, b, c = 1, \dots, p$ for coordinates on the parameter manifold Θ (with the Fisher information metric g_{ab}). Einstein summation is implicit throughout.

2. Samples as geometry: the orthogonal decomposition of $\mathbb{R}^n$

2.1. **The sample point and the mean–deviation splitting.** Fix $n \geq 2$. A sample $(x_1, \dots, x_n)$ is a point

$$\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n.$$

The *mean line* is the one-dimensional subspace

$$L = \{x \in \mathbb{R}^n : x^1 = x^2 = \cdots = x^n\},$$

and its orthogonal complement is the *deviation hyperplane*

$$W = L^\perp = \{v \in \mathbb{R}^n : \textstyle\sum_i v^i = 0\}, \qquad \dim W = n - 1.$$

The orthogonal projection onto L sends each sample point to its componentwise mean:

$$\pi_L(x)^i = \bar{x}, \qquad \bar{x} = \tfrac{1}{n} \textstyle\sum_i x^i.$$

The residual is the *deviation vector*

$$d^i = x^i - \bar{x} \in W.$$

Proposition 2.1 (Pythagorean decomposition)**.**

$$\delta_{ij}(x^i - m)(x^j - m) = n(\bar{x} - m)^2 + \delta_{ij}\, d^i d^j = n(\bar{x} - m)^2 + n\hat{\sigma}^2,$$

where $\hat{\sigma}^2 = \frac{1}{n}\,\delta_{ij}\, d^i d^j$*. This is just the Pythagorean theorem for the decomposition* $\mathbb{R}^n = L \oplus W$*.*

The geometry is simple: the point $(m, \ldots, m) \in L$ is the "true centre" on the mean line, $\pi_L(\mathbf{x}) = (\bar{x}, \ldots, \bar{x})$ is the closest point on L to $\mathbf{x}$, and d^i is the residual in W. The squared distance from $\mathbf{x}$ to the true centre splits into a component along L and a component in W.

Remark 2.2 (From flat to curved)**.** It is worth writing the Pythagorean decomposition once more, emphasizing the *metric* that measures distances:

$$\delta_{ij}(x^i - m)(x^j - m) = n(\bar{x} - m)^2 + \delta_{\alpha\beta}\, \xi^\alpha \xi^\beta,$$

where δ_{ij} is the flat Euclidean metric on sample space and ξ^α ($\alpha = 1, \ldots, n{-}1$) are coordinates on W. The notation is deliberate: in section 5 the *same pattern*—a metric tensor, an orthogonal decomposition, a Pythagorean splitting—reappears on *parameter space*, where the metric is the Fisher information $g_{ab}(\theta)$ and the space is genuinely curved. The flat metric δ_{ij} is the prototype; the information metric g_{ab} is the real thing.

2.2. **The Gaussian measure and orthogonal invariance.** Under the model $x_i \overset{\text{iid}}{\sim} \mathcal{N}(m, \sigma^2)$, the joint density is

$$f(\mathbf{x} \mid m, \sigma) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\Big(-\frac{\delta_{ij}(x^i - m)(x^j - m)}{2\sigma^2}\Big).$$

This density depends on $\mathbf{x}$ only through $\delta_{ij}(x^i - m)(x^j - m)$, so it is *invariant under rotations that fix the true centre* $(m, \ldots, m)$. By the Pythagorean decomposition, it factors as

$$f(\mathbf{x} \mid m, \sigma) = \underbrace{\frac{\sqrt{n}}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{n(\bar{x} - m)^2}{2\sigma^2}\Big)}_{g(\bar{x}|m,\sigma)} \cdot \underbrace{\frac{1}{(2\pi\sigma^2)^{(n-1)/2}} \exp\Big(-\frac{\delta_{ij}\, d^i d^j}{2\sigma^2}\Big)}_{h(\mathbf{d}|\sigma)}. \tag{1}$$

(The change from $(x_1, \ldots, x_n)$ coordinates to $(\bar{x},\, d^1, \ldots, d^{n-1})$ is an orthogonal transformation, so the Jacobian is unity.)

Theorem 2.3 (Independence of $\bar{x}$ and $\mathbf{d}$)**.** *Under the normal model:*

(a) $\bar{x} \sim \mathcal{N}(m,\, \sigma^2/n)$*,*
(b) $\mathbf{d}$ *is uniformly distributed on the sphere in* W *of squared radius* $\delta_{ij}\, d^i d^j$*, with* $\delta_{ij}\, d^i d^j/\sigma^2 \sim \chi^2_{n-1}$*,*
(c) $\bar{x}$ *and* $\mathbf{d}$ *are independent.*

Proof. The factorization (1) exhibits the density as a product of a function of $\bar{x}$ alone and a function of $\mathbf{d}$ alone. Since $\bar{x}$ and $\mathbf{d}$ are images under orthogonal projections onto complementary subspaces of a Gaussian vector, they are independent. The marginal distributions follow from the standard theory of Gaussian projections. □

This is Fisher's "representative point" argument from [1], stripped to its algebraic core. He states it as: "the distributions of $\bar{x}$ and σ_2 are therefore perfectly independent."

3. Comparing estimators of scale

Fisher [1] asks: *which function of the deviation vector* $\mathbf{d}$ *best estimates* σ*?* He compares two classical estimators:

$$\hat{\sigma}_2 = \left(\tfrac{1}{n}\,\delta_{ij}\,d^i d^j\right)^{1/2} = \sqrt{\tfrac{1}{n}\sum(x_i-\bar{x})^2} \qquad \text{(root mean square deviation)},$$

$$\hat{\sigma}_1 = \tfrac{1}{n}\sum|x_i-\bar{x}|\cdot\sqrt{\tfrac{\pi}{2}} \qquad \text{(mean absolute deviation, rescaled)}.$$

Both are rescaled to be asymptotically unbiased for σ.

3.1. **The deviation vector lives on a sphere.** Since $\delta_{ij}\,d^i d^j/\sigma^2 \sim \chi^2_{n-1}$, the *radius* $R = (\delta_{ij}\,d^i d^j)^{1/2}$ has a (scaled) chi distribution with $n-1$ degrees of freedom. The *direction* $\hat{d}^i = d^i/R$ is uniform on the unit sphere $S^{n-2} \subset W$, independent of the radius. So any estimator $T(\mathbf{d})$ that depends only on R depends only on $\hat{\sigma}_2$, while any estimator that depends on the direction carries information not captured by the radius.

3.2. **The mean absolute deviation as a directional statistic.** The mean absolute deviation $\hat{\sigma}_1$ depends on the *signs* of the deviations—that is, on which orthant of W the direction $\hat{\mathbf{d}}$ lies in. For $n=4$, Fisher works out the geometry explicitly: the deviation space W is three-dimensional, and the regions where different numbers of deviations are positive form a partition of S^2 into spherical triangles and quadrangles (the faces of a regular tetrahedron inscribed in the sphere, by the symmetry of permuting coordinates).

3.3. **Relative efficiency.**

Theorem 3.1 (Fisher, 1920)**.** *Among estimators of* σ *of the form* $\hat{\sigma}_p \propto \left(\frac{1}{n}\sum|x_i-\bar{x}|^p\right)^{1/p}$ *for* $p>0$*, the asymptotic relative variance is minimized at* $p=2$*. Specifically, writing* $\mathrm{Var}_{rel}(\hat{\sigma}_p)$ *for the asymptotic relative variance:*

p	1	2	3
Var_{rel}	$\dfrac{\pi-2}{2n}$	$\dfrac{1}{2n}$	$\dfrac{15\pi-8}{72n}$

The ratio $\mathrm{Var}_{rel}(p{=}1)/\mathrm{Var}_{rel}(p{=}2) = \pi-2 \approx 1.14$*, so the mean absolute deviation is* 14% *less efficient than the root mean square deviation.*

Remark 3.2. Equivalently: to achieve the same precision as n observations processed through $\hat{\sigma}_2$, one needs $(\pi-2)\,n \approx 1.14\,n$ observations processed through $\hat{\sigma}_1$.

3.4. **Sufficiency: the geometric punchline.** The deepest result of [1] is not the efficiency comparison but the following:

Theorem 3.3 (Sufficiency of $\hat{\sigma}_2$; Fisher 1920)**.** *For normal data with known mean,* $\hat{\sigma}_2^2$ *is a* sufficient statistic *for* σ*. That is: conditional on* $\hat{\sigma}_2$*, the distribution of the full deviation vector* $\mathbf{d}$ *does not depend on* σ*.*

Equivalently: conditional on $\delta_{ij}\,d^i d^j$*, the direction* $\hat{d}^i$ *is uniform on* S^{n-2}*, regardless of* σ*.*

Proof. The density of $\mathbf{d}$ in $W \cong \mathbb{R}^{n-1}$ is $h(\mathbf{d}\mid\sigma) \propto \sigma^{-(n-1)}\exp(-\delta_{ij}\,d^i d^j/2\sigma^2)$. Writing $R = (\delta_{ij}\,d^i d^j)^{1/2}$ for the radial coordinate and $\hat{d}^i = d^i/R$ for the direction, the density in polar coordinates $(R,\hat{\mathbf{d}})$ becomes

$$\frac{1}{\sigma^{n-1}}\,e^{-R^2/2\sigma^2}\;R^{n-2}\,\mathrm{d}R\;\mathrm{d}\omega(\hat{\mathbf{d}}),$$

where $\mathrm{d}\omega$ is the uniform measure on S^{n-2}. The angular part factors off and does not involve σ. By the Fisher–Neyman factorization theorem, R (equivalently $\hat{\sigma}_2$) is sufficient. $\square$

Remark 3.4 (Sufficiency as a fibre bundle)**.** The radius map $\pi\colon W\setminus\{0\} \to \mathbb{R}_{>0}$, $\pi(\mathbf{d}) = (\delta_{ij}\, d^i d^j)^{1/2}$, is a submersion whose fibres are spheres S_t^{n-2}. On each fibre the conditional measure is the unique rotationally invariant probability (surface measure), independent of σ. Sufficiency has a clean fibre-bundle reading: the parameter governs only the *base* direction (the radial distribution), while the *fibre* direction is ancillary noise.

Corollary 3.5. *For a given value of $\hat{\sigma}_2$, the conditional distribution of $\hat{\sigma}_1$ is independent of σ. Therefore $\hat{\sigma}_1$ provides no additional information about σ beyond what $\hat{\sigma}_2$ already contains.*

Fisher writes this as: "*The whole of the information respecting σ, which a sample provides, is summed up in the value of σ_2.*" This is arguably the first appearance of the concept of sufficiency in the literature.

The geometric picture is clean:

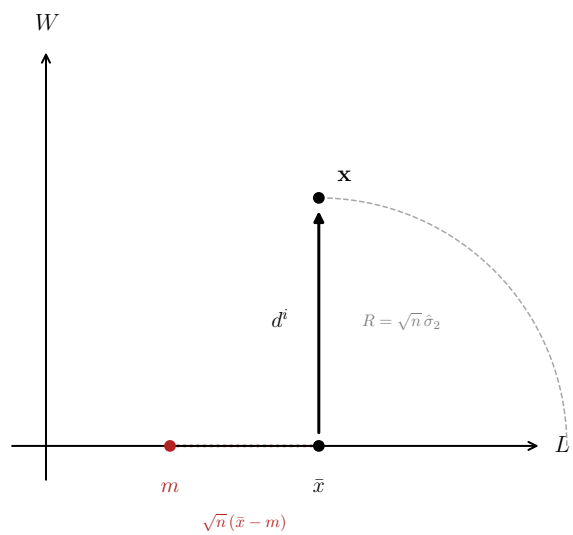

FIGURE 1. The orthogonal decomposition $\mathbb{R}^n = L \oplus W$. The sample $\mathbf{x}$ projects onto L at $\bar{x}$ (capturing the mean) and has deviation $d^i \in W$ (capturing dispersion). The radius $R = \sqrt{n}\,\hat{\sigma}_2$ is sufficient; the direction on the dashed arc is ancillary.

The sample $\mathbf{x}$ decomposes into a point on L (capturing the mean) and a vector in W (capturing dispersion). The distribution of $\mathbf{d}$ in W is isotropic Gaussian, so it is rotationally symmetric: only the radius $R = (\delta_{ij}\, d^i d^j)^{1/2}$ carries information about σ. The direction $\hat{d}^i$ is pure noise. Since $\hat{\sigma}_1$ depends on $\hat{d}^i$ while $\hat{\sigma}_2$ depends only on R, the former is strictly inferior.

4. The correlation coefficient: geometry in $\mathbb{R}^{2n}$

The orthogonal decomposition of $\mathbb{R}^n$ that made sufficiency visible for the scale parameter has an exact analogue for bivariate data: the sample point now lives in $\mathbb{R}^{2n}$, and an angular coordinate on the deviation sphere encodes the correlation. The new wrinkle is that the induced metric on the parameter ρ is *curved*, so the natural coordinate ρ is no longer geodesic.

Let $(x_1, x_1'), \ldots, (x_n, x_n')$ be n pairs drawn from a bivariate normal population with means m, standard deviation σ (common to both variates), and correlation ρ.

4.1. **The sample space and its decomposition.** The full sample is a point

$$\mathbf{w} = (x_1, \ldots, x_n,\, x_1', \ldots, x_n') \in \mathbb{R}^{2n}.$$

The joint density is

$$f(\mathbf{w}) = \frac{1}{(2\pi\sigma^2)^n\,(1-\rho^2)^{n/2}} \exp\!\left(-\frac{1}{2\sigma^2(1-\rho^2)} \sum_{i=1}^{n} \big[(x_i-m)^2 - 2\rho(x_i-m)(x_i'-m) + (x_i'-m)^2\big]\right).$$

Fisher's geometric approach: represent the sample as a point in $\mathbb{R}^{2n}$ and decompose via orthogonal projections. Introduce the statistics

$$\bar{x} = \tfrac{1}{2n}\sum(x_i + x_i'), \qquad \mu^2 = \tfrac{1}{2n}\sum\big[(x_i-\bar{x})^2 + (x_i'-\bar{x})^2\big],$$
$$r = \frac{\sum(x_i-\bar{x})(x_i'-\bar{x})}{\sum(x_i-\bar{x})^2 + \sum(x_i'-\bar{x})^2} \cdot 2.$$

Here $\bar{x}$ is the grand mean, μ is the pooled standard deviation, and r is the sample (intraclass) correlation.

The key factorization is: in the $(2n-1)$-dimensional deviation space perpendicular to the grand mean direction, the representative point lies on a sphere of radius $\mu\sqrt{2n}$. Its position on that sphere is specified by a single *angular coordinate* θ related to r by

$$r = \cos 2\theta.$$

Proposition 4.1 (Sampling distribution of r)**.** *Fisher* [2] *distinguishes two cases.*

Interclass (product-moment) correlation. *Each pair (x_i, x_i') has its own mean and variance estimated separately. The density of r from n pairs is*

$$p_{\text{inter}}(r \mid \rho) = \frac{n-2}{\pi}\,(1-\rho^2)^{(n-1)/2}\,(1-r^2)^{(n-4)/2} \int_0^\infty \frac{\mathrm{d}t}{(\cosh t - \rho r)^{n-1}}. \tag{2}$$

Intraclass correlation. *The two variates play symmetric roles (e.g. siblings in the same family): a common mean and a common variance are assumed, so two fewer parameters are estimated from the data. Rotating to sum and difference variables $u_i = (x_i + x_i')/\sqrt{2}$, $v_i = (x_i - x_i')/\sqrt{2}$, the sum of squares $S_u = \sum(u_i - \bar{u})^2$ has $n-1$ degrees of freedom (one lost to the grand mean), while $S_v = \sum v_i^2$ retains all n (since $\mathbb{E}[v_i] = 0$ is known). This asymmetry gives the density*

$$p_{\text{intra}}(r \mid \rho) = C_n\,(1+\rho)^{n/2}\,(1-\rho)^{(n-1)/2}\,(1+r)^{(n-3)/2}\,(1-r)^{(n-2)/2}\,(1-\rho r)^{-(2n-1)/2}, \tag{3}$$

where $C_n = \Gamma\!\left(\frac{2n-1}{2}\right) \Big/ \left(2^{(2n-3)/2}\,\Gamma\!\left(\frac{n-1}{2}\right)\Gamma\!\left(\frac{n}{2}\right)\right)$. Note the asymmetric *exponents on $(1+r)$ and $(1-r)$, reflecting the different degrees of freedom $n-1$ and n; the interclass formula* (2)*, by contrast, is symmetric in $(1-r^2)$.*

Both distributions are supported on $(-1, 1)$ and are highly skewed *when ρ is far from zero: they pile up against the boundary $r = \pm 1$. In the large-n limit the two formulas agree to leading order, but for small samples the distinction matters—a point Fisher emphasizes with extensive numerical tables.*

The integral representation (2) arises from integrating out the "radial" variable (the pooled variance) from the joint distribution on the sphere, then integrating over the remaining angular variables.

4.2. **Why r is a bad coordinate.** The distribution (2) is a disaster for practical inference:

- Its shape changes with ρ: symmetric when $\rho = 0$, violently skewed near $\rho = \pm 1$.
- Its variance changes with ρ: roughly proportional to $(1-\rho^2)^2$.
- The "probable error" (i.e. interquartile range) is an unreliable summary because the distribution is not even approximately symmetric.

Fisher recognized that these pathologies stem from using r as a coordinate on the parameter space $(-1, 1)$. The cure is to find a coordinate in which the sampling distribution is approximately normal with constant variance.

5. The z-transformation and the information metric

5.1. **Fisher's transformation.**

Definition 5.1 (Fisher's z-transformation)**.**

$$z = \operatorname{arctanh}(r) = \tfrac{1}{2}\log\frac{1+r}{1-r}, \qquad \zeta = \operatorname{arctanh}(\rho) = \tfrac{1}{2}\log\frac{1+\rho}{1-\rho}. \tag{4}$$

Theorem 5.2 (Fisher, 1921)**.** *Under the transformation* (4)*:*

(a) The sampling distribution of z is approximately $\mathcal{N}\big(\zeta,\ \frac{1}{n-3}\big)$, with the approximation improving rapidly as n grows.

(b) The distribution is approximately invariant in shape*: it is nearly the same curve centred at ζ, for all values of ζ.*

(c) The bias of z as an estimator of ζ is of order $1/(n-1)$, a correction that can be applied when precision is needed.

5.2. **Why arctanh? The information metric as a Riemannian tensor.** Fisher's choice of $z = \operatorname{arctanh}(r)$ may seem like nothing more than an inspired guess. But from a modern standpoint it is the *unique natural coordinate* on $(-1, 1)$, dictated by the Fisher information metric—which is a genuine Riemannian metric tensor on the parameter manifold.

Definition 5.3 (Fisher information metric)**.** For a parametric family $\{p(\cdot \mid \theta) : \theta \in \Theta\}$ with parameters θ^a $(a = 1, \dots, d)$, the *Fisher information metric* is the rank-$(0, 2)$ tensor

$$g_{ab}(\theta) = \mathbb{E}\bigg[\frac{\partial \log p(X \mid \theta)}{\partial \theta^a}\ \frac{\partial \log p(X \mid \theta)}{\partial \theta^b}\bigg] = -\mathbb{E}\bigg[\frac{\partial^2 \log p(X \mid \theta)}{\partial \theta^a\, \partial \theta^b}\bigg]. \tag{5}$$

It is symmetric, positive definite, and defines a Riemannian geometry on the parameter space Θ.

Remark 5.4 (Metric intuition)**.** The metric g_{ab} measures *statistical distinguishability*: $\mathrm{d}s^2 = g_{ab}\,\mathrm{d}\theta^a\,\mathrm{d}\theta^b$ is the squared "information distance" between nearby parameter values. For the correlation parameter, $g_{11}(\rho)$ blows up near $\rho = \pm 1$ because distributions with nearby correlation values close to ± 1 produce very different-looking samples—they are far apart in the information metric, even though they are close in the naïve Euclidean distance $|\,\mathrm{d}\rho|$. The z-transformation compensates for this stretching.

5.2.1. *The one-parameter case.* For the correlation parameter ρ alone (fixing m and σ), the parameter manifold is $\Theta = (-1, 1)$ with metric component

$$g_{11}(\rho) = \frac{n\,(1+\rho^2)}{(1-\rho^2)^2} \ \approx\ \frac{n}{(1-\rho^2)^2} \quad \text{(leading order)}. \tag{6}$$

This is the Fisher information for the full sample of n pairs.

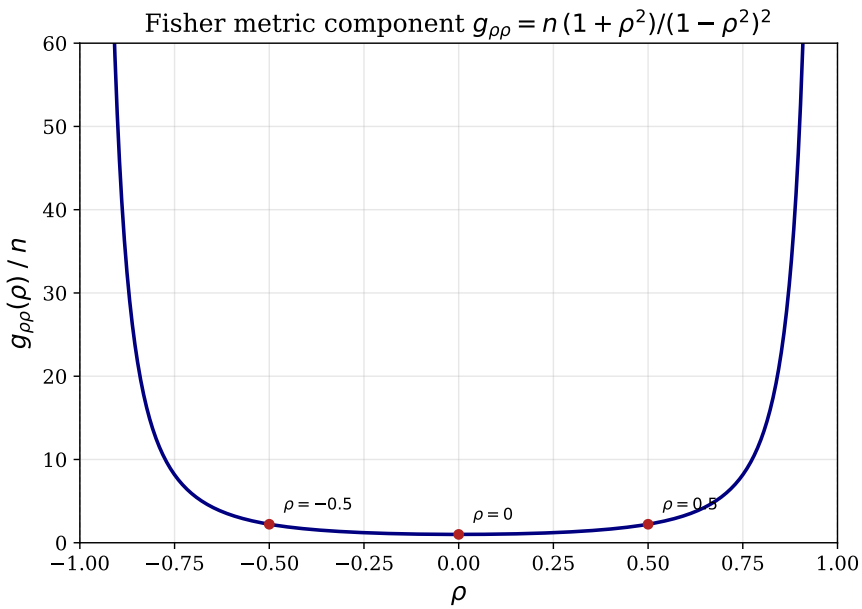

FIGURE 2. The Fisher metric component $g_{11}(\rho)/n = (1+\rho^2)/(1-\rho^2)^2$ on $(-1,1)$. The divergence near $\rho = \pm 1$ reflects the increasing statistical distinguishability of nearby correlations near the boundary. The z-transformation compensates for this stretching.

5.2.2. *The Christoffel symbol and geodesic equation.* In one dimension, the (unique) Christoffel symbol of the Levi-Civita connection is

$$\Gamma^1{}_{11} = \frac{1}{2}\, g^{11}\, \frac{\partial g_{11}}{\partial \rho} \;=\; \frac{2\rho}{1-\rho^2} \quad \text{(leading order)}, \tag{7}$$

and the geodesic equation on $(-1,1)$ reads

$$\ddot{\rho} + \Gamma^1{}_{11}\, \dot{\rho}^2 = 0 \qquad \Longrightarrow \qquad \ddot{\rho} + \frac{2\rho}{1-\rho^2}\, \dot{\rho}^2 = 0. \tag{8}$$

The Christoffel symbol encodes *how bad ρ is as a coordinate*: it is small near $\rho = 0$ (where ρ is approximately geodesic) and diverges near $\rho = \pm 1$ (where ρ is wildly non-geodesic).

5.2.3. *The z-transform as a coordinate change.* Now define $\zeta = \operatorname{arctanh}(\rho)$. The metric transforms as a rank-$(0,2)$ tensor:

$$g'_{11} = \left(\frac{\partial \rho}{\partial \zeta}\right)^2 g_{11} = (1-\rho^2)^2 \cdot \frac{n(1+\rho^2)}{(1-\rho^2)^2} = n(1+\rho^2) \;\approx\; n. \tag{9}$$

The squared Jacobian $(\,\mathrm{d}\rho/\,\mathrm{d}\zeta)^2 = (1-\rho^2)^2$ exactly cancels the $(1-\rho^2)^{-2}$ in g_{11}: *this cancellation is the entire content of the z-transformation.* In the coordinate ζ:

- the metric is approximately constant: $g'_{11} \approx n$,
- the Christoffel symbol approximately vanishes: $\Gamma'^1{}_{11} = \tanh\zeta\ \operatorname{sech}^2\zeta\,/\,(1+\tanh^2\zeta) \approx 0$,
- the geodesic equation linearizes: $\ddot{\zeta} = 0$,
- the variance of the sampling distribution is constant: $\mathrm{Var}(z) \approx 1/(n-3)$.

These are four manifestations of the *same approximate geometric fact*: ζ is the *approximate* geodesic (arclength) coordinate on $(-1,1)$. The residual factor $(1+\tanh^2\zeta)$ in g'_{11} prevents all four from being exact; they become exact in the leading-order metric $g_{11} \approx n/(1-\rho^2)^2$.

Corollary 5.5. *The z-transformation is the* approximate *arclength parametrization of the Fisher information metric on $(-1,1)$: the transformed metric $g'_{11} = n(1+\tanh^2\zeta)$ is approximately constant (ranging between n and $2n$), so $(-1,1)$ with the information metric is approximately isometric to $(\mathbb{R},\, n\,\mathrm{d}\zeta^2)$. The residual factor $(1+\tanh^2\zeta)$ prevents an exact isometry. The leading-order expressions throughout this section drop the factor $(1+\rho^2)$, incurring a relative error of $\rho^2/(1+\rho^2)$: below 10% for $|\rho|<0.3$ but reaching 45% at $|\rho|=0.9$.*

5.2.4. *Non-constancy of the metric.* A one-dimensional Riemannian manifold is intrinsically flat (the Riemann tensor vanishes identically), so there is no curvature in the usual sense. But we can still ask: *how far is ρ from being a geodesic coordinate?* A natural measure is

$$\kappa = -\frac{1}{2\,g_{11}}\,\frac{\mathrm{d}^2}{\mathrm{d}\rho^2}\big(\log g_{11}\big) \;\approx\; -\frac{2}{n} \quad \text{(leading order, at } \rho=0). \tag{10}$$

When $\kappa = 0$, the metric is constant and ρ is already an arclength coordinate; the normal approximation is exact. The fact that $\kappa \approx -2/n$ tells us that ρ is a "bad" coordinate whose distortion shrinks with sample size: as $n\to\infty$, $\kappa\to 0$ and the normal approximation to the sampling distribution of r improves—precisely because the metric becomes approximately flat in the original coordinate.

5.2.5. *The multi-parameter metric.* The full bivariate normal model with equal means and variances has three free parameters $\theta^a = (m,\sigma,\rho)$. The Fisher information metric on the full parameter space is

$$g_{ab} = n\begin{pmatrix} \dfrac{2}{\sigma^2(1+\rho)} & 0 & 0 \\ 0 & \dfrac{4}{\sigma^2} & \dfrac{-2\rho}{\sigma(1-\rho^2)} \\ 0 & \dfrac{-2\rho}{\sigma(1-\rho^2)} & \dfrac{1+\rho^2}{(1-\rho^2)^2} \end{pmatrix}. \tag{11}$$

The off-diagonal entry $g_{23} = -2n\rho/[\sigma(1-\rho^2)] \neq 0$ tells us that *the σ-direction and the ρ-direction are not orthogonal* in the information metric. (The negative sign reflects the fact that increasing ρ effectively reduces the "independent information per pair" available for estimating σ.) Changing ρ while holding σ fixed moves you along a path that is not a geodesic of the full metric—a fact invisible in the one-parameter treatment.

Under the coordinate change $\rho \to \zeta = \operatorname{arctanh}(\rho)$, the $(3,3)$-component transforms exactly as in (9), while the off-diagonal coupling g_{23} transforms to $g'_{23} = -2n\rho/[\sigma(1-\rho^2)]\cdot(1-\rho^2) = -2n\rho/\sigma$, which remains nonzero but is now bounded.

Remark 5.6 (When the full metric matters)**.** For the single-parameter story of [2], the one-dimensional slice (6) suffices. But the full metric (11) becomes essential in at least two settings.

Joint confidence regions. The inverse g^{ab} of the Fisher information is the asymptotic covariance of the MLE $(\hat m, \hat\sigma, \hat\rho)$. A joint 95% confidence ellipsoid in (σ,ρ)-space (profiling out m) is

$$g_{ab}\,(\hat\theta^a - \theta^a)(\hat\theta^b - \theta^b) \;\le\; \chi^2_{2,\,0.95},$$

where a, b range over the $\{\sigma, \rho\}$ block. Because $g_{23} \neq 0$, this ellipsoid is *tilted*: the principal axes are not aligned with the (σ, ρ) coordinate axes. Reporting separate marginal intervals for σ and ρ misses the correlation between the estimates.

Second-order bias. The Christoffel symbols ${\Gamma^a}_{bc}$ of the full 3-parameter metric enter the $O(1/n)$ bias correction for the MLE (Efron, 1975). The z-transformation zeroes out the one-dimensional Christoffel symbol (approximately), but the multi-parameter corrections from g_{23} persist.

The index notation scales seamlessly: the same formula $g'_{ab} = (\partial\theta^c/\partial\theta'^a)\,(\partial\theta^d/\partial\theta'^b)\,g_{cd}$ works whether a runs over one index or five.

Remark 5.7 (Triple coincidence in one dimension)**.** In the one-parameter exponential subfamily obtained by fixing m and σ at known values, the arclength coordinate $\zeta = \operatorname{arctanh}\rho$ simultaneously achieves three things: (i) it is the approximate arclength of the Levi-Civita metric (the Christoffel symbol approximately vanishes), (ii) it is the *e-affine coordinate* for the exponential connection of Amari–Nagaoka (the natural parameter of this subfamily), and (iii) it is the variance-stabilizing transformation. In dimension $d > 1$, these three notions give *different* coordinate systems, and the geometry becomes genuinely richer. See Amari and Nagaoka [5] for the full theory of dual connections on statistical manifolds.

This is the modern explanation: the z-transform linearizes the information geometry. In any one-parameter exponential family, the variance-stabilizing transformation that makes the sampling distribution approximately $\mathcal{N}(\zeta,\, 1/I)$ with constant I is precisely the arclength of the Fisher metric. Fisher discovered this specific instance a decade before the general theory existed.

5.3. **Hyperbolic geometry.** There is an even more concrete geometric picture. The interval $(-1, 1)$ with the metric $\mathrm{d}s^2 = \mathrm{d}\rho^2/(1-\rho^2)^2$ is a model of the *hyperbolic line* $\mathbb{H}^1$. The map $\rho \mapsto \operatorname{arctanh}(\rho)$ is the standard isometry from the Poincaré-disc model to the "velocity" (or Klein) parametrization.

In this language: *the natural geometry on the space of correlations is hyperbolic*, and the z-transform is nothing but a change to geodesic (arc-length) coordinates. The fact that the sampling distribution of r is skew and has non-constant variance reflects the curvature of the hyperbolic metric in the ρ-coordinate; passing to z flattens the geometry.

The following table summarizes the geometric objects in both coordinate systems (leading-order expressions, dropping the residual factor $1 + \rho^2$):

Geometric object	**ρ-coordinate**	**ζ-coordinate**
Metric g_{11}	$\dfrac{n}{(1-\rho^2)^2}$	n
Christoffel symbol ${\Gamma^1}_{11}$	$\dfrac{2\rho}{1-\rho^2}$	0
Geodesics	$\rho(t) = \tanh(at+b)$	$\zeta(t) = at + b$
Distance $d(\rho_1, \rho_2)$	$\sqrt{n}\,\lvert\operatorname{arctanh}\rho_1 - \operatorname{arctanh}\rho_2\rvert$	$\sqrt{n}\,\lvert\zeta_1 - \zeta_2\rvert$

The key point is that geodesic distance in the ρ-coordinate is *not* proportional to $|\rho_1 - \rho_2|$. The information-geometric distance between $\rho = 0$ and $\rho = 0.5$ is $\sqrt{n} \cdot 0.549$, while the distance between $\rho = 0.5$ and $\rho = 0.9$ is $\sqrt{n} \cdot 0.923$. Correlations that "look close" in the ρ-scale are "far apart" in the information metric when they are near ± 1. This is why confidence

intervals for ρ that are symmetric in ρ are wrong: they ignore the curvature. The z-transform produces confidence intervals that are symmetric in the geodesic coordinate.

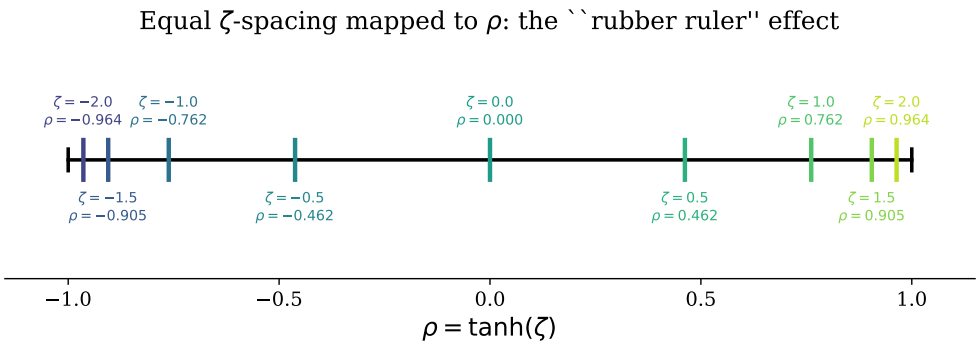

FIGURE 3. Equal steps in ζ produce shrinking steps in $\rho = \tanh(\zeta)$ near the boundary. Each interval represents the same information distance $\Delta\zeta = 0.5$. The compression visualizes the Fisher metric: a "rubber ruler" that stretches near $\rho = \pm 1$.

6. Worked examples

We pause to show how the geometric machinery translates into the kind of calculations Fisher himself performed.

Example 6.1 (Testing a single correlation)**.** A biologist observes $r = 0.45$ from $n = 20$ pairs and wishes to test $H_0\colon \rho = 0$ against $H_1\colon \rho \neq 0$.

Step 1 (transform). $z = \operatorname{arctanh}(0.45) = \frac{1}{2}\log\frac{1.45}{0.55} \approx 0.4847$.

Step 2 (standard error). Under H_0, $\zeta_0 = 0$ and $\operatorname{Var}(z) \approx 1/(n-3) = 1/17$, so $\operatorname{SE}(z) = 1/\sqrt{17} \approx 0.2425$.

Step 3 (test statistic). $Z = (z - \zeta_0)/\operatorname{SE}(z) = 0.4847/0.2425 \approx 2.00$.

At the 5% level the critical value is 1.96, so the null is (just barely) rejected. The entire calculation rests on the geometric fact that ζ is the approximate arclength coordinate: in this coordinate the sampling distribution is nearly $\mathcal{N}(\zeta,\, 1/(n{-}3))$, so ordinary Gaussian inference applies.

Example 6.2 (Combining correlations from independent studies)**.** Three independent studies report

$$r_1 = 0.30\ (n_1 = 25), \quad r_2 = 0.55\ (n_2 = 40), \quad r_3 = 0.40\ (n_3 = 15).$$

We seek a pooled estimate of the common ρ.

Transform. $z_k = \operatorname{arctanh}(r_k)$: $z_1 \approx 0.310$, $z_2 \approx 0.618$, $z_3 \approx 0.424$.

Pool. Each z_k has approximate variance $1/(n_k - 3)$, so the inverse-variance weighted mean is

$$\bar{z} = \frac{\sum w_k z_k}{\sum w_k}, \qquad w_k = n_k - 3.$$

Numerically: $w_1 = 22$, $w_2 = 37$, $w_3 = 12$; $\sum w_k = 71$; $\bar{z} = (22 \cdot 0.310 + 37 \cdot 0.618 + 12 \cdot 0.424)/71 \approx 0.490$.

Back-transform. $\hat{\rho} = \tanh(\bar{z}) \approx \tanh(0.490) \approx 0.454$, with 95% CI for ζ: $0.490 \pm 1.96/\sqrt{71} \approx (0.257,\, 0.722)$, giving a CI for ρ: $(\tanh 0.257,\ \tanh 0.722) \approx (0.251,\ 0.618)$.

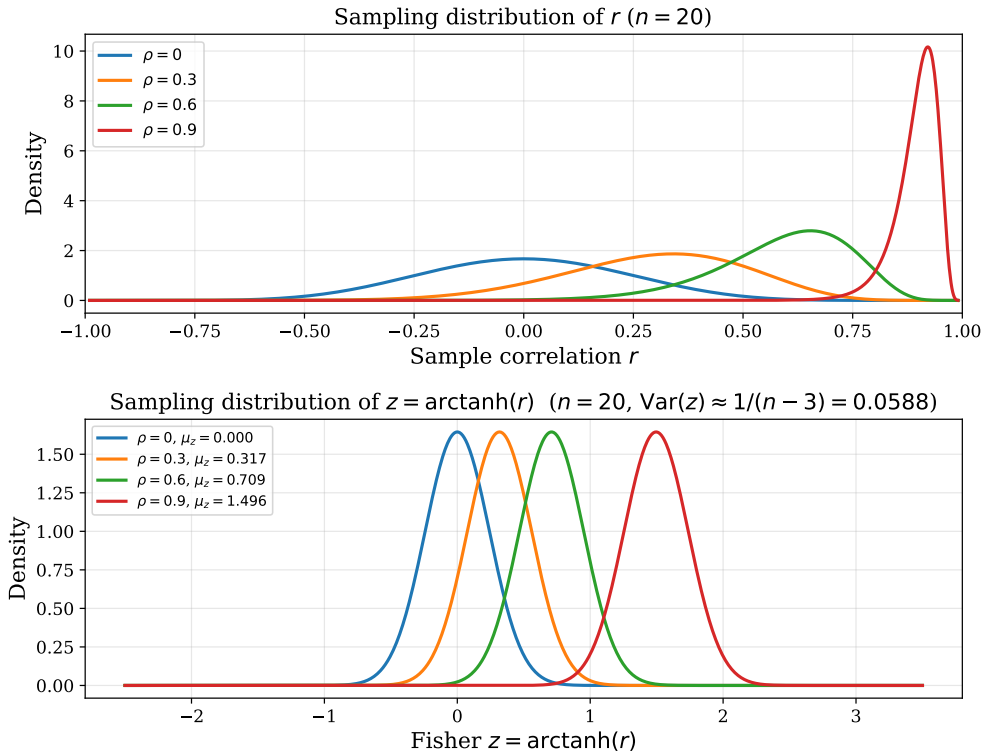

FIGURE 4. Sampling distributions of the correlation coefficient ($n = 20$). **Top**: the distribution of the sample correlation r is increasingly skewed and compressed against the boundary as ρ grows. **Bottom**: in the geodesic coordinate $z = \mathrm{arctanh}(r)$, the distributions are approximately Gaussian with constant variance $\mathrm{Var}(z) \approx 1/(n-3)$, regardless of the true correlation.

The pooling is a *Euclidean average in geodesic coordinates*: because ζ is the approximate arclength coordinate, the information-weighted mean has a clean probabilistic interpretation.

Example 6.3 (Naïve vs. geodesic confidence interval)**.** A sample of $n = 10$ pairs gives $r = 0.80$. Compare two 95% confidence intervals for ρ.

Naïve (symmetric in ρ). A textbook shortcut gives $\mathrm{SE}(r) \approx (1 - r^2)/\sqrt{n} = 0.36/\sqrt{10} \approx 0.114$, yielding $0.80 \pm 1.96 \cdot 0.114 = (0.577,\ 1.023)$. This extends past the boundary $\rho = 1$—an absurdity caused by ignoring the curvature.

Geodesic (symmetric in ζ). $z = \mathrm{arctanh}(0.80) \approx 1.099$; $\mathrm{SE}(z) = 1/\sqrt{7} \approx 0.378$; CI for ζ: $(1.099 \pm 0.741) = (0.358,\ 1.840)$; back-transform: $(\tanh 0.358,\ \tanh 1.840) \approx (0.343,\ 0.951)$.

The geodesic interval is asymmetric in ρ (wider below r, narrower above) and respects the boundary—exactly as the information geometry predicts. The asymmetry reflects the fact that the Fisher metric stretches distances near $\rho = \pm 1$: a step of $\Delta\rho = 0.1$ near $\rho = 0.9$ is informationally larger than the same step near $\rho = 0$.

7. Sufficiency and the exponential family viewpoint

Both papers, viewed from the modern standpoint of exponential families, tell the same story. The index notation makes the underlying duality visible.

7.1. **The normal location-scale family.** An exponential family density takes the canonical form

$$f(\mathbf{x} \mid \theta) = h(\mathbf{x}) \exp\bigl(\eta^a(\theta)\, T_a(\mathbf{x}) - A(\eta)\bigr),$$

where η^a are the *natural parameters* and T_a are the *sufficient statistics*, paired by contraction. The index placement reflects genuine tangent/cotangent duality: η^a are coordinates on Θ (tangent type), while the *expectation parameters* $\theta_a = \partial A/\partial\eta^a = \mathbb{E}[T_a]$ are the dual (cotangent type) coordinates. (Note the subscript placement, distinguishing these from the parameter coordinates θ^a used elsewhere.) The Fisher information metric mediates between them:

$$\mathrm{d}\theta_a = g_{ab}\, \mathrm{d}\eta^b \qquad \text{(infinitesimal Legendre duality).}$$

In full, the metric is the Hessian of the log-partition function,

$$g_{ab}(\eta) = \frac{\partial^2 A}{\partial\eta^a\, \partial\eta^b} = \mathrm{Cov}(T_a,\, T_b). \tag{12}$$

This equation encodes three facts at once: the metric is symmetric ($g_{ab} = g_{ba}$), positive definite (covariance of a non-degenerate family), and connects the geometry of parameter space to the statistics of sufficient statistics.

For the normal location-scale family, $\eta^1 = m/\sigma^2$, $\eta^2 = -1/(2\sigma^2)$, and the sufficient statistics are $T_1 = \sum x_i,\ T_2 = \sum x_i^2 = \delta_{ij}\, x^i x^j$. The pairing is

$$\eta^a T_a = \frac{m}{\sigma^2} \sum x_i - \frac{1}{2\sigma^2}\, \delta_{ij}\, x^i x^j.$$

Equivalently, the pair $(\bar{x},\, \hat{\sigma}_2^2)$ is jointly sufficient for (m, σ). No other function of $\mathbf{x}$ carries any additional information about the parameters—this is the content of theorem 3.3.

7.2. **The bivariate normal family.** Similarly, for n bivariate normal pairs the exponential family has *five* natural parameters (two means, two variances, one covariance) and five sufficient statistics. When means and variances are treated as nuisance parameters and marginalized out, the remaining information about ρ is captured entirely by r.

The complication is that the *conditional distribution of r given the nuisance sufficient statistics* is what we actually observe, and it is this distribution that proposition 4.1 describes. The z-transformation then provides the right coordinate on the marginal parameter.

8. The joint distribution for $n = 4$: spherical geometry

Fisher treats the case $n = 4$ in detail in [1] because the deviation space W is three-dimensional and can be visualized.

8.1. **The tetrahedron on the sphere.** For $n = 4$, the deviation vector $\mathbf{d} \in W \cong \mathbb{R}^3$ encodes three degrees of freedom. The constraint $d_1 + d_2 + d_3 + d_4 = 0$ means the vertices of a regular tetrahedron in $\mathbb{R}^4$ project to a regular tetrahedron in W. The hyperplanes $d_i = 0$ partition $S^2 \subset W$ into 14 regions:

- 6 spherical quadrangles (where two deviations are positive and two negative),
- 8 spherical triangles (where one or three deviations are positive).

The mean absolute deviation $\hat{\sigma}_1$ is constant on each quadrangle (by symmetry) and varies across the triangles. Its conditional distribution given $\hat{\sigma}_2 = R/2$ (where $R = (\delta_{ij}\, d^i d^j)^{1/2}$) is determined by the uniform distribution on S^2 restricted to these regions. Fisher computes the resulting frequency curve and shows it has a discontinuity—a feature that disappears in

the large-n limit but illustrates concretely why the mean absolute deviation is a "coarser" statistic than the root mean square deviation.

8.2. **The angle between estimators.** In the case $n = 4$, Fisher derives a relationship between $\hat{\sigma}_1$ and $\hat{\sigma}_2$ through a polar angle θ on S^2:

$$\cos\theta = \sqrt{\frac{2}{\pi}} \cdot \frac{\hat{\sigma}_1}{\hat{\sigma}_2},$$

i.e. the ratio $\hat{\sigma}_1/\hat{\sigma}_2$ is a cosine on the deviation sphere. The conditional distribution of $\hat{\sigma}_1$ given $\hat{\sigma}_2$ becomes the distribution of an angular coordinate on a sphere—a purely geometric quantity independent of any parameters.

9. Probability versus likelihood

Fisher [2] contains what is likely the first published articulation of the distinction between probability and likelihood:

> "So defined, probability and likelihood are quantities of an entirely different nature. Probability is transformable as a differential element; ... likelihood ... is not a differential element, and is incapable of integration. Again the *mode* of a frequency curve may be arbitrarily shifted by transforming the variable ... but no transformation can alter the likelihood of any suggested value of ρ."

In the language of differential geometry, the distinction is a statement about *transformation type*. The *score function*

$$s_a(\theta) = \frac{\partial \log p(x_{\text{obs}} \mid \theta)}{\partial \theta^a}$$

is a *covector* (1-form) on parameter space Θ: under reparametrization $\theta'^b = \phi^b(\theta)$, it transforms as $s'_b = (\partial\theta^a/\partial\theta'^b)\, s_a$. The *Fisher information metric* is the expected outer product of this covector,

$$g_{ab}(\theta) = \mathbb{E}\big[s_a(\theta)\, s_b(\theta)\big],$$

which is manifestly a rank-$(0,2)$ tensor on Θ.

The *likelihood function* $L(\theta) = p(x_{\text{obs}} \mid \theta)$ is a *scalar*: it assigns a number to each point of Θ with no Jacobian, no volume element, no density transformation. Its maximum is therefore a *point* in Θ, invariant under reparametrization. The maximum likelihood estimate solves $s_a(\hat{\theta}) = 0$—a tensorial equation (covector equals zero)—so its solution is coordinate-independent.

9.1. **Fisher's invariance argument.** Fisher illustrates the distinction with the correlation coefficient itself. The sampling distribution of r (proposition 4.1) is a genuine probability density: if we reparametrize by $z = \operatorname{arctanh}(r)$, the density transforms as

$$p(z \mid \zeta)\ \mathrm{d}z = p\big(\tanh z \mid \tanh\zeta\big)\ \operatorname{sech}^2 z\ \mathrm{d}z.$$

The Jacobian $\operatorname{sech}^2 z$ changes both the shape and the location of the mode. In particular, the mode of $p(r \mid \rho)$ as a function of r is *not* at $r = \rho$ (the distribution is skewed), and the mode shifts further under reparametrization. Fisher's objection is that any quantity whose "most probable value" depends on which coordinate system one happens to use cannot be a fundamental notion.

The likelihood $L(\rho) = p(r_{\text{obs}} \mid \rho)$, viewed as a function of the *parameter* ρ with r_{obs} fixed, carries no Jacobian: $L(\rho) = L(\tanh\zeta)$ has the same numerical values, and hence the same

maximum, in either coordinate. The MLE $\hat{\rho} = r_{\text{obs}}$ (for the interclass case) is a coordinate-invariant statement, while "the most probable value of ρ" is not.

9.2. **Geometric taxonomy of statistical objects.** *Probability*, by contrast, transforms as a *density*: if $r = g(\rho)$, then $p(r)\,\mathrm{d}r = p(g(\rho))\,|g'(\rho)|\,\mathrm{d}\rho$. The mode of a density depends on the Jacobian and hence on the parametrization. We can now classify the objects that appear in inference by their transformation type:

Object	**Type on Θ**	**Transforms with...**
Likelihood $L(\theta)$	scalar (function)	nothing
Score $s_a = \partial_a \log L$	covector (1-form)	$\partial\theta/\partial\theta'$
Fisher metric g_{ab}	$(0,2)$-tensor	$(\partial\theta/\partial\theta')^2$
Posterior density $\pi(\theta)$	density (top form)	$\lvert\det(\partial\theta/\partial\theta')\rvert$

Fisher's point is exactly this: the likelihood is an intrinsic object on the parameter manifold (Θ, g_{ab}), while a posterior density requires a *volume form* (a prior), which is additional structure beyond the Riemannian geometry. The information metric provides one natural volume form—the Riemannian density $\sqrt{\det g_{ab}}\,\mathrm{d}\theta$, yielding Jeffreys' prior—but this is a choice, not a theorem.

For the correlation coefficient, Jeffreys' prior is $\pi_J(\rho) \propto \sqrt{g_{11}(\rho)} = \sqrt{(1+\rho^2)}/(1-\rho^2)$, which concentrates mass near $\rho = \pm 1$—reflecting the information-geometric fact that these boundary regions are "large" in the metric. A uniform prior on ρ implicitly assumes a flat geometry that the data contradict; a uniform prior on ζ assumes a different geometry. The likelihood does not care: $L(\rho) = L(\tanh\zeta)$ has the same maximum in either coordinate.

10. Summary of key ideas

Fisher's language	**Modern language**	**Paper**
Representative point	$\mathbf{x} \in \mathbb{R}^n$, metric δ_{ij}	[1]
$\mathrm{S}(x-\bar{x})^2 = n\sigma_2^2$	$\delta_{\alpha\beta}\,\xi^\alpha\xi^\beta$, squared residual in W	[1]
Frequency surface	Joint density / measure	[1, 2]
$\bar{x}$ and σ_2 independent	$L \perp W$ decomposition of Gaussian	[1]
σ_2 is a *sufficient* estimate	Sufficient statistic (Fisher–Neyman)	[1]
Minimum-variance derivate	Cramér–Rao bound g^{ab}	[1]
Fisher information	Metric tensor $g_{ab}(\theta)$	[1, 2]
Sampling curve of r	Distribution (2)	[2]
$z = \frac{1}{2}\log\frac{1+r}{1-r}$	$\operatorname{arctanh}(r)$: arclength of g_{11}, $\Gamma^1{}_{11} \approx 0$	[2]
Probable error	Interquartile range / std. error	[2]
Most likely value $\neq$ probability	$L(\theta)$ is a scalar, $s_a = \partial_a \log L$ is a covector	[2]

11. Afterword: what Fisher started

These two papers, read together, inaugurate three of the central themes of twentieth-century statistics:

1. **Sufficiency and data reduction.** The insight that the isotropic Gaussian in W concentrates all parametric information in the radius leads directly to the Fisher–Neyman factorization theorem (1922/1935) and ultimately to the theory of exponential families.
2. **Information geometry.** The z-transformation is the first example of a variance-stabilizing / information-metric-flattening reparametrization: the coordinate ζ in which the metric tensor g_{11} becomes approximately constant and the Christoffel symbol ${\Gamma^1}_{11}$ approximately vanishes. The general principle—that the sampling distribution becomes approximately Gaussian in the geodesic coordinate of the Fisher metric—was formalized by Efron (1975, 1978) and is central to modern differential-geometric statistics. The full theory of dual connections on statistical manifolds (Amari and Nagaoka [5]) extends these ideas to multi-parameter families, where the metric g_{ab} is a genuine matrix and the geometry becomes irreducibly tensorial.
3. **Likelihood inference.** The distinction between probability and likelihood, drawn explicitly in [2], is precisely the distinction between a density (which transforms with a Jacobian) and a scalar function on the parameter manifold (which does not). The score $s_a = \partial_a \log L$ is a covector, the Fisher metric is its expected outer product $g_{ab} = \mathbb{E}[s_a\, s_b]$, and the MLE solves the tensorial equation $s_a(\hat{\theta}) = 0$. These ideas underpin Fisher's subsequent development of maximum likelihood (1922), ancillarity, and conditional inference.

That all three themes appear, in embryonic but unmistakable form, in the work of a 30-year-old writing about astronomical error measurement and botanical correlations is a remarkable testament to the power of geometric thinking in statistics.

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